Leeds CITY COUNCIL

Pedestrian Crossing Site Assessment Guidelines

2016

PEDESTRIAN CROSSING SITE ASSESSMENT GUIDELINES

Introduction

The previous assessment framework for the provision of pedestrian crossing facilities has been revised and expanded to provide a stronger framework and method for ensuring that the full range of relevant factors are considered during the assessment process. The revised assessment matrix and guidelines for use was approved in June 2016 by the Chief Officer (Highways and Transportation). Separate supplementary advice and guidance for officers in the use of the methodology have also been prepared and form an Annex to the approved framework.

As per previous reviews of the assessment process this guidance continues to take full regard to the current government guidance Local Transport Note (LTN) 1/95 however the use of the previous numerical PV^2 criteria no longer forms part of the formal assessment. While traffic volumes, speeds and pedestrian demand remain the key factors, the new assessment framework gives greater emphasis to the potential community benefits where a crossing would assist in overcoming severance; whilst at the same time considering any potential dis-benefits, such as loss of passing trade, parking, access to premises and public transport. Implications for road safety (impact on speeds and potential casualties) and community interest are also considered. Additional difficulties experienced by children, older people and disabled people when crossing the road likewise are formalised in the assessment process.

A scoring system has been developed as part of the new framework to enable an impartial assessment to be made by weighing the different factors, and the thresholds required for a formal/ informal crossing have been benchmarked against the sites currently meeting the criteria for provision. The technical considerations included in this document as to what type of measure is the most appropriate have largely remained unchanged.

Appendix 1

ASSESSOR……

PEDESTRIAN CROSSING ASSESSMENT FRAMEWORK

DAY/DAY/TIME……

SITE……

WEATHER & ROAD CONDITIONS……..

Section 1: Site Assessment

SCORE	-3	-2		-1	0	1	2	3	Total
Impact on the local community		A worsening of condition in both i. Access to frontage property ii. Restrictions on waiting		A worsening of conditions in either: i. Access to frontage property ii. Restrictions on waiting	10 properties or less benefiting	Whole Street of up to 50 properties benefiting	Local neighbourhood of up to 200 properties benefiting	Benefits extending beyond the local area	
Crossing impact on the locality/ businesses	A worsening of conditions in **ALL** of: i) Access to premises made more difficult ii) Passing trade removed iii) Restrictions on waiting iv) Noise/Visual Pollution	A worsening of conditions in any **TWO** of: i) Access to premises made more difficult ii) Passing trade removed iii) Restrictions on waiting iv) Noise/Visual Pollution		A worsening of conditions in **ONE** of: i) Access to premises made more difficult ii) Passing trade removed iii) Restrictions on waiting iv) Noise/Visual Pollution	No real impact but maybe a couple of properties benefiting at most (commercial/ industrial)	A parade of 15 shops or business properties benefiting	A district centre benefiting	A town centre benefiting	
Public Interest					First request in 3 years	Two independent requests in last 12 months	Significant community interest with multiple representations		
Traffic Speed Assessment					Mean speeds within prescribed limit; the speed limit itself is 20mph (preferred)	Mean speeds exceed the limit by up to 10%, or the speed limit is 30mph or above	Mean speeds exceed the limit by up to 20%, or the speed limit is 40mph or above	Mean speeds exceed the limit by up to 30%, or the speed limit is 50mph or more	
Highway Assessment	I.	II. **Use Section 2 – Highway Assessment score**							
Road Safety History	III.	IV. **Use Section 3 – Road Safety History score**							
Traffic/ Pedestrian Surveys	V.	VI. **Use Section 4 – Traffic/Pedestrian score**							
							VII. TOTAL SCORE		

Section 2: Highway Assessment

Road character: Two way single carriageway, Dual Carriageway, etc	Type of Road Single carriageway	Road Classification B	Direction of flow (2 way)
Carriageway width: *Between islands or central reserve for dual carriageways	Overall Width	Lane 1*.	Lane 2*.
Other road features (presence of alternative crossings, refuges islands, traffic calming, TROs etc):-			
Other road factors (adjacent junctions, accesses etc):-			
Frontage (✓ any)	Shops	Residential	School
Other (hospital, day centre etc.):-			
Bus services/stops proximity:- yes			
Visual check of crossing opportunities (circle one): (0) Very easy - no difficulty within a few seconds (0) Easy - short wait up to 30 seconds (1) Moderate difficulty - wait of up to one minute (2) Difficult - more than a one minute wait (3) Very difficult - long wait of two minutes or more (3) Impossible - after waiting several minutes for an opportunity Judgement should be based on normal walking pace WITHOUT having to walk fast or run to cross in safety.			

Section 3: Road safety history

Accidents:	5 year period from 2010 to 2015		
Severity	slight	serious	fatal
Adult pedestrian			
Child pedestrian			
Others			
Other factors:-			
-1	**0**	**1**	**2**
Risk potential increased	No effect on safety	Risk potential reduced	Some accident savings possible

Note: Recorded for 50 metres either side of study site.

Section 4: Traffic/Pedestrian Surveys

Traffic/Ped surveys:	12 hours	Busiest hour	Second busiest hour
Flow:-	_____to_____	_____to_____	_____to_____
All vehicles			
Adult pedestrians (all)			
Child pedestrians			
Elderly people			
Other relevant groups 1. 2.			
Other details:-			
Speed Limit	85 percentile	Average (mean)	

Pedestrian volumes per hour at busiest hours:

25 – 50 = 1 point, 50 – 75 = 2 points, >75 = 3 points.

High volume of child/ elderly pedestrians + 1 point

Conclusions/ recommendations:

Pedestrian Crossing Assessment Framework – notes for guidance

1) The purpose of this assessment framework is to ensure that the Council fulfils the requirements of LTN 1/95 “The assessment of pedestrian crossings” when considering requests for pedestrian crossings. The framework considers the difficulty of crossing and existing pedestrian demand as well as overall benefits and dis-benefits of the potential provision for both pedestrians and local residents and businesses, as well as impact on road safety.
2) This approach is a development of the previous process approved by the Director of Highways and Transportation in 2002 (revised 2006) and has been benchmarked against previously approved crossings.
3) The first approach to all requests is an initial site inspection followed by a desk top study of the available accident and traffic data. As a rule this will be followed up by a 12 hour pedestrian and traffic volumes survey. The survey will help determine the busiest times for both pedestrians and traffic and this in turn will inform the best periods for site observation.
4) The site visit should note the following
    a) Any community facilities that are present (shops, library, school, community centre, pubs, bus stops, surgeries, Post Office, etc)
    b) Current parking arrangements (driveways, on-street parking)
    c) Presence of any passing trade (foot and motorised)
    d) Any pedestrian desire lines/ attractors
    e) Any observed crossing difficulties and contributing factors (age, disability, highway characteristics, parking)
    f) Any nearby features that facilitate crossing
5) The appropriate information needs to be entered into the assessment sheet, including data from the desktop study (speeds, accidents, pedestrian and vehicles volumes and pedestrian profile).
6) For sites which receive the score of >8 a formal crossing is recommended – the exact type of the facility to be determined by the nature of the road, traffic and pedestrian flows and vehicular speeds, as per Pedestrian Crossing Site Assessment Guidelines.
7) In making recommendations, the assessor should be seeking to examine the most effective and economic means of ensuring that the observed volume of pedestrian traffic can cross the road in safety. In essence, the objective is to provide measures which allow pedestrians the time they need to cross, either by a formal crossing, or where numbers or traffic flow does not justify it, the appropriate informal measures such as refuge islands, promontories etc

Appendix 2
PEDESTRIAN CROSSING TYPOLOGY AND SELECTION CRITERIA

Indicative Framework Score	**Previous reference PV^2 value**	**Guidelines for appropriate crossing provision**	**Typical site characteristics and road conditions**
Category A Score of 8 and above	>0.75 (busiest 2 hours, all pedestrians)	Puffin crossing will generally be preferred for the busiest sites. Used at school or crossing patrol sites only where significant other pedestrian movements exist.	Very busy road where traffic speed >35 mph 85th percentile. Typically traffic flows will exceed 1000 vehicles per hour and over 70 pedestrian movements in busiest hours, or there will be an indication of supressed demand. At some sites there may be a record of pedestrian injuries. Pedestrian waiting time will generally exceed 1 minute. For sites are at the lower end of speed and traffic range zebra crossings will be preferred.
Category B Score of 8 and above	0.6 – 0.85 (busiest 2 hours, all pedestrians)	Zebra crossing will generally be preferred at these quieter sites. In some instance other informal measures may be recommended. There will be evident community benefit in either overcoming severance or improving access to local facilities and services.	Medium trafficked road with flows typically over 700 vehicles and where traffic speed <35 mph 85th percentile. Pedestrian flows will typically exceed 40 in the busiest hours and should exceed those on adjacent sections of road by at least 3:1 thereby demonstrating a clear desire line. Most sites unlikely to have a pattern of pedestrian casualties. Waiting times up to 30 seconds and occasionally exceeding 1 minute. Some sites at the higher end of the range may be best suited to Puffin crossing control.
Category C 4-8	<0.6 (busiest 2 hours, all pedestrians)	Informal measures to assist those having difficulty crossing the road. At SCP sites package of measures to assist warden or as part of a school travel initiative may be appropriate.	Lightly trafficked road where flows usually <600 v.p.h. provide ample and frequent gaps in traffic. No discernible pedestrian desire line nor usually a pattern of pedestrian road injuries. Minimal delay crossing road within 30 seconds of reaching it. Exceptionally a formal crossing may be justified where there is additional crossing difficulty due to the road layout and/ or significant presence of vulnerable pedestrians, and where informal measures are not feasible.

ANNEX A

Pedestrian Crossing Assessment Framework

During the course of each year requests for the provision of new pedestrian crossings are received from members of the public and elected members. In order to prepare recommendations for a programme for the installation of new crossings, all such requests are investigated and the results collated and analysed. A formal process for assessing new pedestrian crossing site was first approved by the former Highways and Transportation Committee in 1996, following the publication of LTN 1/95 “The assessment of pedestrian crossings”.

Further developments to the assessment process were made following reviews in 2002 and 2006, progressively moving the process away from the purely numerical criteria (peak hour pedestrian flow multiplied by the square of peak hour traffic flows i.e.) to more qualitative rather than quantitative focussed approach (PV^2 is not now part of the formal process but optionally can help to benchmark the findings). This increases the place of external factors such as the presence of attractors (local facilities, schools, bus stops etc) that the community were trying to access, road safety history and the presence of those pedestrians for whom crossing the road may present additional difficulties: children, older people and disabled people.

This multi-criteria framework introduces a score-based system allowing the above factors to be quantified in order to achieve objective outcomes of the assessment process, building on examples of best practice from neighbouring local authorities and long-running experience in assessing requests. The revised process establishes thresholds for the introduction of a formal or informal facility based on community benefit, road safety, observed difficulty of crossing (especially as experienced by children, older people and disabled people) and benefits for the locality as well as any potential dis-benefits, that have been carefully benchmarked against outcomes of previously assessed requests.

Scores between 4 and 8 indicate some degree of crossing difficulty which can be eased by informal measures (for example refuges, junction narrowing or build-outs). Scores above 8 indicate that a formal facility (a Zebra or a signal controlled crossing) should be considered. Higher scores, arising from higher traffic speeds and volume, greater crossing difficulty and road safety record, may indicate the need for a higher-end facility (signal controlled crossing). However, the choice of the facility will be predominantly rest on the road and traffic characteristics weighted to the pedestrian demand and waiting times and subject to a feasibility, engineering and road safety assessment.

Whilst signal controlled crossing are generally more appropriate on busier and faster roads, zebra crossings can provide safe facilities where speeds are lower and can achieve reduced pedestrian delay and convenience. Overall, where used appropriately, zebra crossings have achieved safety records just as good as equivalent light controlled crossings.

Typical site characteristics (demand and traffic conditions) for a variety of formal controlled and uncontrolled crossings are set out in Appendix 2 to the guidance.Final draft, 2016

ANNEX B

Guidance on assessing the site requirements for pedestrian crossings for new developments

This advice sets out the approach for considering the basis on which proposals for the inclusion of freestanding crossings within new developments should be considered.

The overall principle for new developments is that the Council should be satisfied that the proposal is justified in terms of forecast future demand, having regard for existing observed demand in line with the overall principles and volumetric guidelines (traffic and pedestrian flows) set out in the appendices below.

In order to satisfy this requirement the following approach should be taken:

i) At the location where a crossing is being considered, the baseline traffic and pedestrian flows at the site prior to development should be established by an appropriate survey. This information will provide an initial indication as to what type of crossing provision would normally be considered appropriate. This will also enable an informed decision to be made in terms of any pre-existing pedestrian demand prior to considering the impact of the proposed development.

ii) Once the baseline position is established the site Transport Assessment (if one is required) should be used to make evaluate the likely future pedestrian traffic at the proposed crossing site. This estimate should reflect typical peak and off-peak hourly conditions. In order to reach a comparative traffic and pedestrian demand figures in line with the guidelines, the development demand figure should be combined with the existing baseline figures to provide an indicative assessment score (similar to the purposed of the former PV^2 criteria) for the proposed crossing location. This information can then be used to determine the scale of provision that should be sought as part of the development.

iii) Where the proposed development does not have a Transport Assessment, an estimate of the potential pedestrian demand and its temporal distribution should be made. It is expected that this will be based on size and type of development and the anticipated patterns of movement from the site onto the highway. Again this estimate should be combined within existing observed data for pedestrian and traffic movements to produce an indicative assessment score for the location.

There are also instances where pedestrian crossing demand may change as a result of other local authority proposals which do not of themselves trigger a planning application or a development control process. The school reorganisation programme is one such example where adjacent schools may be amalgamated and this change the distribution and demand for pedestrian facilities on the local network. In such instances the principles outlined above should still apply. That is to say that the

baseline traffic and pedestrian figures should be established and an estimate of the likely change in demand arising from the proposal should be overlaid with the existing pedestrian demand and traffic values i.e. for a school this would initially be achieved by plotting and comparing school catchment plots and following this with a survey of families to determine desire lines and a robust forecast of pedestrian movements.

In line with the technical guidance contained within the Appendices 1 and 2 other external factors to the site should be taken into account in the same way as they would for any non-development related site.

None of this advice supersedes or substitutes for the wider assessment of the planning application.

Final draft, 2016